

26. The quadratic mean of two real numbers  $x$  and  $y$  equals  $\sqrt{(x^2+y^2)/2}$ .

By computing the arithmetic and quadratic means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Solution: ①  $x=3, y=4$  A.M.  $= (x+y)/2 = (3+4)/2 = 3.5 = \sqrt{12.25}$

Q.M.  $= \sqrt{(x^2+y^2)/2} = \sqrt{(3^2+4^2)/2} = \sqrt{12.5}$

②  $x=4, y=6$  A.M.  $= (4+6)/2 = 5 = \sqrt{25}$  Q.M.  $= \sqrt{(4^2+6^2)/2} = \sqrt{26}$

③  $x=2, y=8$  A.M.  $= (2+8)/2 = 5 = \sqrt{25}$  Q.M.  $= \sqrt{(2^2+8^2)/2} = \sqrt{34}$

④  $x=6, y=8$  A.M.  $= (6+8)/2 = 7 = \sqrt{49}$  Q.M.  $= \sqrt{(6^2+8^2)/2} = \sqrt{50}$

⑤  $x=5, y=5$  A.M.  $= (5+5)/2 = 5 = \sqrt{25}$  Q.M.  $= \sqrt{(5^2+5^2)/2} = 5$

Claim: For any two arbitrary real numbers  $x$  and  $y$ ,  $A.M. = (x+y)/2 \leq Q.M. = \sqrt{(x^2+y^2)/2}$  for any two

Proof: Suppose, for the sake of contradiction,  $\frac{x+y}{2} > \sqrt{\frac{x^2+y^2}{2}}$

arbitrary real numbers  $x$  and  $y$ .  
 $\therefore \frac{(x+y)^2}{4} > \frac{x^2+y^2}{2} \Rightarrow x^2+2xy+y^2 > 2x^2+2y^2 \Rightarrow x^2-2xy+y^2 < 0 \Rightarrow (x-y)^2 < 0$

Since,  $(x-y)^2$  is a real number,  $(x-y)^2 \geq 0$  (A Fundamental Property of Reals)  
 This is a contradiction.

$\therefore$  For any two arbitrary real numbers  $x$  and  $y$ ,  $\frac{x+y}{2} \leq \sqrt{\frac{x^2+y^2}{2}}$   $\square$

27. Write the numbers  $1, 2, \dots, 2n$  on a blackboard, where  $n$  is an odd integer. Pick any two of the numbers,  $j$  and  $k$ , write  $|j-k|$  on the board and erase  $j$  and  $k$ . Continue this process until only one integer is written on the board.